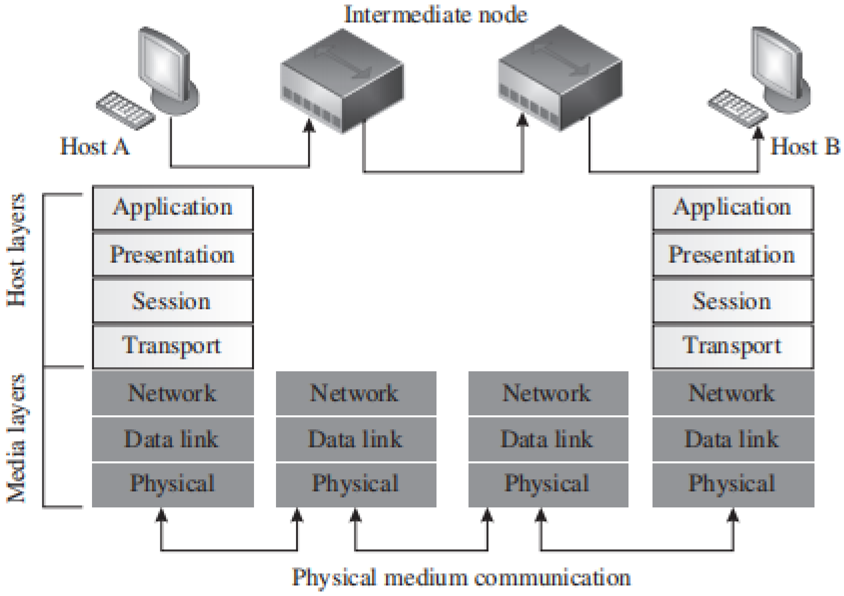
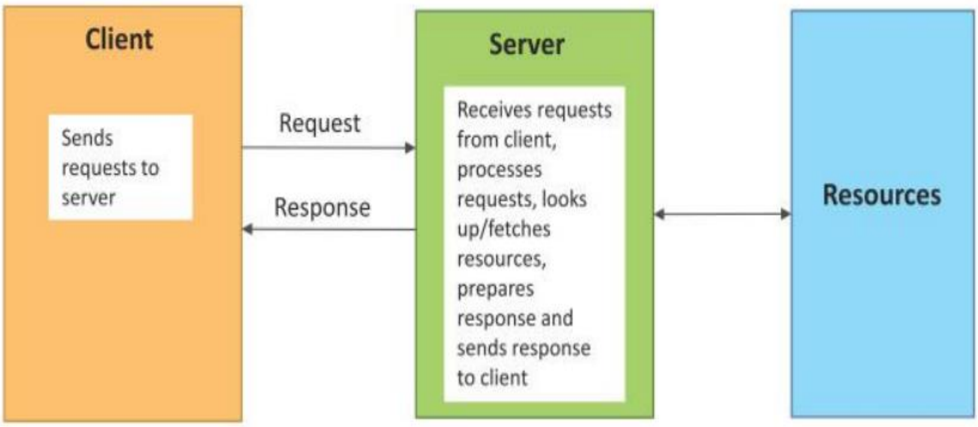
Introduction to IoT

Module 1

2 Marks	
1	**Define networking** **Networking** refers to linking of computers and communication network devices , which interconnect through a network and are separated by unique device
2	**List the layers present in internet protocol suite** **Layers of internet protocol suite** 1. Link layer 2. Internet layer 3. Transport layer 4. Application layer
3	**Define IoT** **IOT(Internet of Thingd)** refers to the network of physical objects embedded with sensors , software and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet

5 Marks	
1	A. **List the layers of OSI model** **B. Draw the figure to show the networked communication between the two hosts following the OSI model** **A. Layers of OSI Model** 1. Physical Layer 2. Data Link Layer 3. Network Layer 4. Transport Layer 5. Session Layer 6. Presentation Layer 7. Application Layer

	B. Networked communication between two hosts following the OSI model 
2	**Describe the working of Request-Response IoT communication Model Request Response Communication Model**  *Request-Response :* Request-Response is a communication model in which the client sends requests to the server and the server responds to the requests. When the server receives a request ✓ it decides how to respond, fetches the data, ✓ retrieves resource representations, ✓ prepares the response, ✓ and then sends the response to the client. Request-Response model is a stateless communication model and each request-response pair is independent of others
10 Marks	
1.	**List the types of Physical topologies and Compare these in terms of their features, advantages, and disadvantages.**

	There are four types of physical topologie 1. Star 2. Mesh 3. Bus 4. Ring

Topology	Feature	Advantage	Disadvantage
Star	Point-to-point	Cheap; ease of installation; ease of fault identification	Single point of failure; traffic visible to network entities
Mesh	Point-to-point	Resilient against single point of failures; scalable; traffic privacy and security ensured	Costly; complex connections
Bus	Point-to-multipoint	Ease of installation; cheap	Length of backbone cable limited; number of hosts limited; hard to localize faults
Ring	Point-to-point	Ease of installation; cheap; ease of fault identification	Prone to single point of failure

2. **List and explain the components of IoT networking node**

1. IoT router
2. IoT LAN
3. IoT WAN
4. IoT gateway
5. IoT proxy.

IoT Node: Devices in an IoT LAN with sensors, processor, and communication unit. They collect data and communicate with other nodes or through a gateway.

IoT Router: A network device that routes data between IoT devices. It ensures proper data flow and can also act as a gateway.

IoT LAN: A local network connecting IoT devices within a small area (like a building) using short-range communication. May or may not be connected to the Internet.

IoT WAN: A wide network that connects multiple LANs over large distances and provides Internet access.

IoT Gateway: Connects IoT LAN to WAN/Internet. It forwards data between networks and works mainly at the network layer.

	IoT Proxy: Works at the application layer to provide security and manage communication between IoT devices and external systems.

Module 2

2 Marks	
1	**Define sensor resolution.** The smallest change in the measurable quantity that a sensor can detect is referred to as the resolution of a sensor.
2	**List the different ways on which sensors are broadly classified.** Sensors are classified based on: **i.** power requirements, **ii.** sensor output, and **iii.** property to be measured.
3	**Depending on the urgency of data processing how are IoT data classified ?** Depending on the urgency of data processing IoT data are classified as **a)** Very time critical **b)** time critical and **c)** normal.
5 Marks	
1	**Explain the following types of actuators** **i. Pneumatic actuators** **ii. Soft actuators** i. **Pneumatic actuators** ● A **pneumatic actuator** works on the compression and decompression of gases. ● It uses compressed air or vacuum to produce linear or rotary motion. ● These actuators convert pressure into force and respond quickly to control signals. ● Small pressure changes can generate large forces. ● Commonly used in applications like valve control (rack and pinion) and pneumatic brakes. ii. **Soft actuators** ● **Soft actuators** are made of elastomeric polymers embedded in flexible materials like cloth, fibers, or paper. ● They work by converting microscopic molecular changes into visible deformation.

	• These actuators play an important role in modern robotics. • They are suitable for handling delicate tasks such as fruit harvesting and robot-assisted surgeries.
2.	**Explain the different types of offload location** The different types of offload location are : 1. Edge 2. Fog 3. Remote server 4. Cloud • **Edge**: Offloading processing to the edge implies that the data processing is facilitated to a location at or near the source of data generation itself. • **Fog**: Fog computing is a decentralized computing infrastructure that is utilized to conserve network bandwidth, reduce latencies, restrict the amount of data unnecessarily flowing through the Internet, and enable rapid mobility support for IoT devices. • **Remote Server**: A simple remote server with good processing power may be used with IoT-based applications to offload the processing from resource constrained IoT devices. • **Cloud**: Cloud computing is a configurable computer system, which can get access to configurable resources, platforms, and high-level services through a shared pool hosted remotely.
10 Marks	
1	**Explain any five main factors governing the IoT design and selection consideration** The critical factors to be considered during the design of IoT devices 1. Size 2. Energy 3. Cost 4. Memory 5. Processing power: 6. I/O rating 7. Add-ons 1. **Size:** The size of a sensor node is important as it affects both form factor and energy consumption. Smaller devices are preferred in IoT applications because they consume less power and can be easily deployed in compact or large-scale environments.

2. **Energy:** Energy requirement is a critical factor in IoT design. Devices with high power consumption need frequent battery replacement, which reduces long-term sustainability, especially in remote or large deployments.
3. **Cost:** The cost of processors and sensors determines how many nodes can be deployed. Lower-cost devices allow higher density deployment, making IoT solutions more efficient and affordable.
4. **Memory:** Memory capacity defines the ability of a device to store and process data locally. Functions like data filtering, storage, and formatting depend on memory, but higher memory increases the overall cost. □
5. **Processing Power:** Processing power decides the complexity of operations a device can perform. It determines the types of sensors supported and the level of data processing that can be done on the device itself.
6. **I/O Rating:** The input-output rating affects compatibility and circuit complexity. Newer processors operate at lower voltages (3.3V), which may require additional circuitry to interface with older components (5V).
7. **Add-ons:** Additional features like ADC, built-in clock, USB, Ethernet, and wireless connectivity enhance the functionality and flexibility of IoT devices, and also help in faster system development

2 **List and explain the characteristics of actuators.**

The characteristics of actuators are

1. Weight:
2. Power Rating:
3. Torque to Weight Ratio:
4. Stiffness and Compliance

Weight: The weight of an actuator affects its application. Heavier actuators are usually used in industrial or fixed systems, while lightweight actuators are preferred in portable devices like drones, vehicles, and home IoT systems. However, some heavy actuators are also used in mobile systems such as aircraft engines and landing gear.

Power Rating: Power rating defines the safe operating limits of an actuator in terms of voltage and current. It is often expressed as a power-to-weight ratio. For example, small servo motors may operate at around 5V and 500 mA, suitable for battery-powered applications.

Torque to Weight Ratio: This ratio indicates the actuator’s efficiency and sensitivity. A higher torque-to-weight ratio means better performance. If the moving part is heavier, the

	ratio decreases for the same power. **Stiffness and Compliance:** Stiffness is the ability of a material to resist deformation, while compliance is the opposite. Stiff actuators provide faster and more accurate responses to load changes, whereas compliant actuators are more flexible but less precise

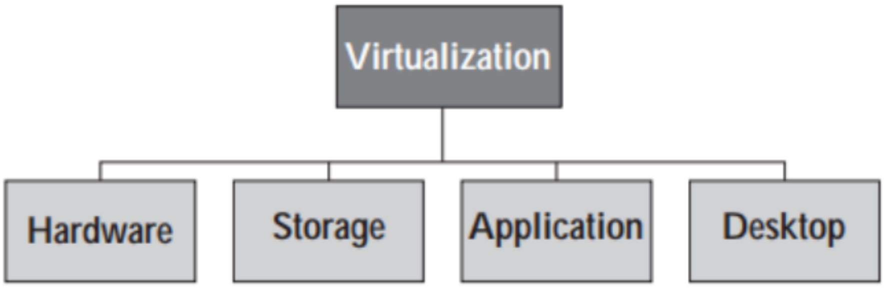
Module 3

	2 Marks
1	**Name the two entities in a cloud computing architecture** Two entities in a cloud computing architecture are 1. end users and 2. Cloud Service provider(CSP)
2	**List the three types of sensors equipped in The TelosB motes** **The TelosB** motes are equipped with three types of sensors: 1. Temperature sensor 2. Humidity sensor **3.** light sensor
3	**What is OpenStack ?** **The OpenStack** is free software, which provides a cloud IaaS to users
	5 marks
1	**Explain the metrics used in Service-Level Agreement(SLA)** The metrics used in Service-Level Agreement (SLA) are : **i.** **Availability:** This metric signifies the amount of time the service will be accessible for the customer **ii.** **Response Time:** The maximum time that will be taken for responding to a customer request is measured by response time. **iii.** **Portability:** This metric indicates the flexibility of transferring the data to another service. **iv.** **Problem Reporting:** How to report a problem, whom and how to be contacted, is explained in this metric. **v.** **Penalty:** The penalty for not meeting the promises mentioned in the SLA.
2	**What are the features of Cloudsim?** **Ans:** The features of cloudsim are

<table>
<tr><td></td><td>1. Using CloudSim, virtualization of server hosts can be done in a simulation.
2. A user is able to allocate virtual machines (VMs) dynamically.
3. It allows users to define their own policies for the allocation of host resources to VMs.
4. It provides flexibility to add or remove simulation components dynamically.
5. A user can stop and resume the simulation at any instant of time</td></tr>
<tr><td colspan="2">10 Marks</td></tr>
<tr><td>1</td><td>What re the features of
A. Green Cloud
B. Open stackA. Ans: Features of GreenCloud

A. Features of GreenCloud
1. GreenCloud is an open-source simulator with user-friendly GUI.
2. It provides the facility for monitoring the energy consumption of the network and its various components.
3. It supports the simulations of cloud network components.
4. It enables improved power management schemes.
5. It allows a user to manage and configure devices, dynamically, in simulation

B. Features of Open stack
1. OpenStack allows a user to create and deploy virtual machines.
2. It provides the flexibility of setting up a cloud management environment.
3. OpenStack supports an easy horizontal scaling: dynamic addition or removal of instances for providing services to multiple numbers of users.
4. This cloud platform allows the users to access the source code and share their code to the community.</td></tr>
<tr><td>2.</td><td>What are the types of Virtualization? Explain
Types of virtualization
1. Hardware Virtualization
2. Storage Virtualization
3. Application Virtualization
4. Desktop Virtualization</td></tr>
</table>



1) Hardware Virtualization:

- This type of virtualization indicates the sharing of hardware resources among multiple users.
- For example, a single processor appears as many different processors in a cloud computing architecture.Different operating systems can be installed in these processors and each of them can work as stand-alone machines

2) Storage Virtualization:

- In storage virtualization, the storage space from different entities are accumulated virtually, and seem like a single storage location.
- Through storage virtualization, a user's documents or files exist in different locations in a distributed fashion. However, the users are under the impression that they have a single dedicated storage space provided to them.

3) Application Virtualization:

- A single application is stored at the cloud end. However, as per requirement, a user can use the application in his/her local computer without ever actually installing the application.
- Similar to storage virtualization, in application virtualization, the users get the impression that applications are stored and executed in their local computer.

4) Desktop Virtualization:

- This type of virtualization allows a user to access and utilize the services of a desktop that resides at the cloud. The users can use the desktop from their local desktop